

Task 1: Nmap Labs Replication

This report documents the replication of all Nmap labs covered in class. Each scan is presented with the command used, the full output, and an explanation of what the results mean from a security perspective. All scans were performed on Kali Linux using Nmap 7.95 on 7 May 2026.

1.1 Identifying the Machine IP — ip address

Before any scanning begins, the attacker must identify their own IP address. The ip address command lists all active network interfaces and their assigned addresses.

Command

```
ip address
```

```
root@kali: ~  
Session Actions Edit View Help  
(root@kali)~  
# ip address  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def  
ault qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
roup default qlen 1000  
    link/ether 08:00:27:a1:0e:82 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 85536sec preferred_lft 85536sec  
    inet6 fd17:625c:f037:2:704b:4dad:d56b:6296/64 scope global temporary dyna  
mic  
        valid_lft 86262sec preferred_lft 14262sec  
    inet6 fd17:625c:f037:2:a00:27ff:fe1:e82/64 scope global dynamic mngtmpad  
dr noprefixroute  
        valid_lft 86262sec preferred_lft 14262sec  
    inet6 fe80::a00:27ff:fe1:e82/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
(root@kali)~
```

Figure 1: ip address output — Kali machine is 10.0.2.15 on eth0

The output shows two interfaces. The loopback (lo) has address 127.0.0.1 and is only used for internal communication. The ethernet interface (eth0) holds the IP 10.0.2.15/24, confirming the machine is connected to the 10.0.2.0/24 subnet. This IP is used as the basis for all subsequent network scans.

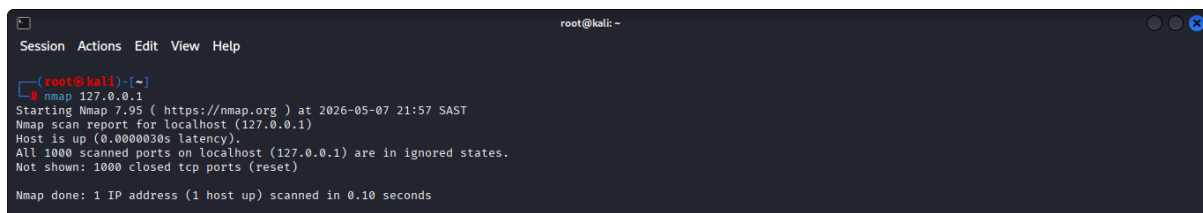
Security implication: Knowing your own IP is the first step in any engagement. It determines what subnet to scan and what address to set as LHOST when configuring reverse shell payloads.

1.2 Basic Default Scan — nmap 127.0.0.1

The simplest Nmap invocation scans the top 1,000 TCP ports of the target. Running it against localhost (127.0.0.1) is a safe first test.

Command

```
nmap 127.0.0.1
```

A terminal window titled 'root@kali: ~' showing the execution of the command 'nmap 127.0.0.1'. The output indicates that Nmap 7.95 was started at 2026-05-07 21:57 SAST. The scan report for localhost (127.0.0.1) shows that the host is up with a latency of 0.0000030s. All 1000 scanned ports are in ignored states, and 1000 closed TCP ports (reset) were not shown. The scan completed in 0.10 seconds.

```
root@kali: ~  
# nmap 127.0.0.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-07 21:57 SAST  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.0000030s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
Nmap done: 1 IP address (1 host up) scanned in 0.10 seconds
```

Figure 2: Basic scan, SYN scan, and full port scan results on localhost

Result: All 1,000 ports are reported closed (reset). The Kali machine has no active listening services. The scan completed in 0.10 seconds, demonstrating how fast Nmap is against a local target with no latency.

Security implication: A clean baseline like this is important. Any open ports found in later scans need to be identified and justified.

1.3 SYN Stealth Scan — nmap -sS

A SYN scan sends SYN packets and analyses responses without completing the full three-way handshake. This makes it harder to detect in logs compared to a full TCP connect scan.

Command

```
nmap -sS 127.0.0.1
```

A terminal window titled 'root@kali: ~' showing the execution of the command 'nmap -sS 127.0.0.1'. The output is identical to the basic scan, showing that all 1000 scanned ports are in ignored states and 1000 closed TCP ports (reset) were not shown. The scan completed in 0.11 seconds.

```
root@kali: ~  
# nmap -sS 127.0.0.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-07 21:58 SAST  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.0000050s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds
```

Fig 3. Basic scan, SYN scan, and full port scan results on localhost

Result (see Figure 2): 1,000 closed ports (reset). Because no services are running, the output is identical to the basic scan. The methodological difference matters in real engagements where stealth is required.

Security implication: SYN scans are the most commonly used scanning method precisely because they leave fewer traces in application logs, though they are still visible to network-level IDS.

1.4 Full Port Scan — nmap -p-

Adding -p- instructs Nmap to probe all 65,535 TCP ports instead of the default top 1,000. This is essential for finding services running on non-standard ports.

Command

```
nmap -p- 127.0.0.1
```

A terminal window titled 'root@kali: ~' showing the execution of the command 'nmap -p- 127.0.0.1'. The output shows that Nmap 7.95 was started at 2026-05-07 21:58 SAST. The scan report for localhost (127.0.0.1) shows that the host is up with a latency of 0.0000030s. All 65535 scanned ports are in ignored states, and 65535 closed TCP ports (reset) were not shown. The scan completed in 0.60 seconds.

```
root@kali: ~  
# nmap -p- 127.0.0.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-07 21:58 SAST  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.0000030s latency).  
All 65535 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 65535 closed tcp ports (reset)  
Nmap done: 1 IP address (1 host up) scanned in 0.60 seconds
```

Fig 4. Comprehensive full port scan (all 65,535 ports) on localhost

Result: All 65,535 ports closed. The scan took 0.60 seconds; six times longer than the default scan, because it checked 65x more ports. On a live server, this would uncover hidden services that a default scan would miss.

Security implication: Administrators who hide services on high port numbers (security through obscurity) are easily discovered by a full port scan.

1.5 Aggressive Scan on Localhost — nmap -A -T4

The -A flag activates OS detection, service version detection, default NSE scripts, and traceroute simultaneously. The -T4 timing template uses aggressive timeouts for faster results.

Command

```
nmap -A -T4 127.0.0.1
```



```
root@kali: ~  
Session Actions Edit View Help  
All 65535 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 65535 closed tcp ports (reset)  
Nmap done: 1 IP address (1 host up) scanned in 0.60 seconds  
  
root@kali: ~  
# nmap -A -T4 127.0.0.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-07 22:00 SAST  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000000s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
Too many fingerprints match this host to give specific OS details  
Network Distance: 0 hops  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 3.11 seconds  
  
root@kali: ~  
# nmap -O 127.0.0.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-07 22:01 SAST  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000000s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
Too many fingerprints match this host to give specific OS details  
Network Distance: 0 hops  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 1.61 seconds  
  
root@kali: ~  
#
```

Figure 5: Aggressive scan and OS-only scan on localhost

Result: All 1,000 ports closed. OS detection returned "Too many fingerprints match" because no open ports were available for fingerprinting. Network Distance 0 hops confirms we scanned ourselves. The scan took 3.11 seconds due to the additional detection overhead.

1.6 Aggressive Scan on Local Network — nmap -A -T4 10.0.2.0/24

The /24 notation tells Nmap to scan all 256 addresses in the 10.0.2.0 subnet, discovering all live hosts and their services simultaneously.

Command

```
nmap -A -T4 10.0.2.0/24
```

```

root@kali: ~
Session Actions Edit View Help
root@kali:~# nmap -s -T4 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-07 22:01 SAST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000073s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.04 seconds

root@kali:~# nmap -s -T4 10.0.2.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-07 22:02 SAST
Nmap scan report for 10.0.2.2
Host is up (0.00175s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
445/tcp   open  microsoft-ds?
MAC Address: 52:54:00:12:35:80 (QEMU virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AT&T BGM210 voice gateway (94%), Oracle Virtualbox Slirp NAT bridge (89%), QEMU user mode network gateway (89%), HP 9100c Digital Sender printer (J3113A) (86%), Minolta Di550 laser printer (85%), NEC SuperScript printer (85%), Cisco SG 500 switch (85%), Yamaha RX-V2007 or RX-V3900 audio receiver (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb2-security-mode:
| 3:111:

```

Figure 8: Network scan discovers 10.0.2.2 with ports 135 and 445 open

```

root@kali: ~
Session Actions Edit View Help
root@kali:~# nmap -s -T4 10.0.2.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-07 22:02 SAST
Nmap scan report for 10.0.2.2
Host is up (0.00175s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
445/tcp   open  microsoft-ds?
MAC Address: 52:54:00:12:35:80 (QEMU virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AT&T BGM210 voice gateway (94%), Oracle Virtualbox Slirp NAT bridge (89%), QEMU user mode network gateway (89%), HP 9100c Digital Sender printer (J3113A) (86%), Minolta Di550 laser printer (85%), NEC SuperScript printer (85%), Cisco SG 500 switch (85%), Yamaha RX-V2007 or RX-V3900 audio receiver (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb2-security-mode:
| 3:111:
|_ Message signing enabled and required
|_ smb2-time:
|_ date: 2026-05-07T20:06:15
|_ start_date: N/A
|_ clock-skew: 3m36s

TRACEROUTE
HOP RTT ADDRESS
1 1.72 ms 10.0.2.2

Nmap scan report for 10.0.2.3
Host is up (0.0038s latency).
Not shown: 999 filtered tcp ports (net-unreach)

```

Figure 7: SMB security details and traceroute for 10.0.2.2

```

root@kali: ~
Session Actions Edit View Help
TRACEROUTE
HOP RTT ADDRESS
1 1.72 ms 10.0.2.2

Nmap scan report for 10.0.2.3
Host is up (0.0038s latency).
Not shown: 999 filtered tcp ports (net-unreach)
PORT      STATE SERVICE      VERSION
53/tcp    open  domain       dnsmasq 2.71
|_ dns-nsid:
|_ bind.version: dnsmasq-2.71
MAC Address: 52:54:00:12:35:80 (QEMU virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter/bridge[general purpose]
Running (3)ST QUES5ING): AT&T embedded (91%), Oracle Virtualbox (89%), Slirp (89%), QEMU (89%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny.gasparovski:slirp cpe:/s:qemu:qemu
Aggressive OS guesses: AT&T BGM210 voice gateway (91%), Oracle Virtualbox Slirp NAT bridge (89%), QEMU user mode network gateway (89%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE
HOP RTT ADDRESS
1 3.01 ms 10.0.2.3

Nmap scan report for 10.0.2.15
Host is up (0.000073s latency).
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 236 IP addresses (3 hosts up) scanned in 26.81 seconds

```

Figure 8: Host 10.0.2.3 (DNS) and 10.0.2.15 (our Kali) — scan summary: 3 hosts up

Result: Three live hosts discovered in 26.81 seconds:

- 10.0.2.2 — Windows host with ports 135 (MSRPC) and 445 (SMB) open. SMB2 message signing is enabled and required. OS guessed as Windows.
- 10.0.2.3 — DNS gateway running dnsmasq 2.71 on port 53. Identified as a VirtualBox NAT bridge device.
- 10.0.2.15 — Kali machine with no open ports, confirming our localhost results.

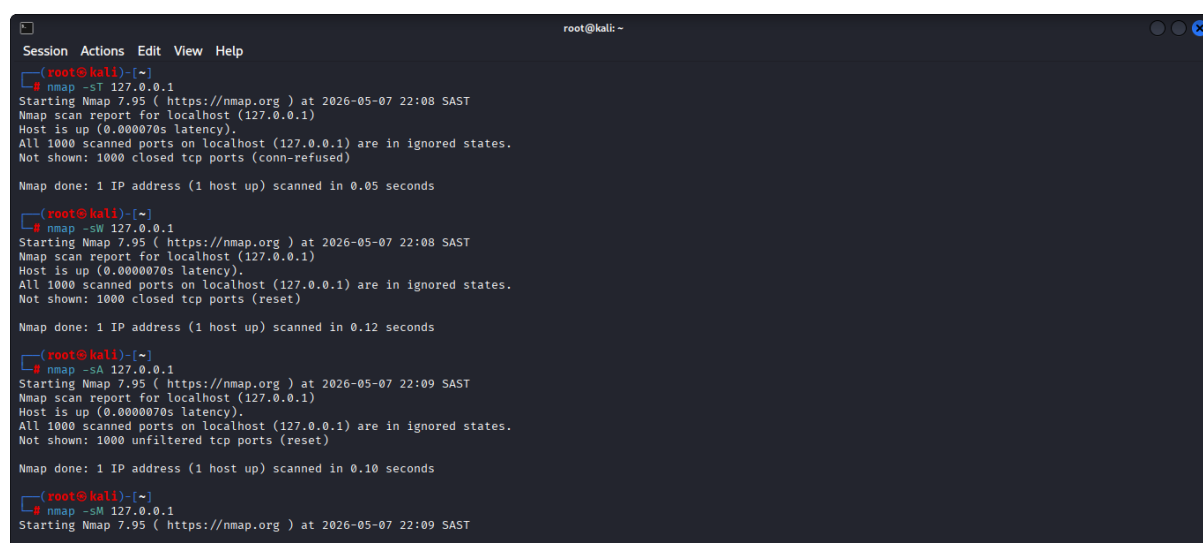
Security implication: A subnet scan like this maps an entire network in seconds. Exposed SMB on port 445 is historically the vector for exploits like EternalBlue (MS17-010). Finding port 53 exposed on a gateway could enable DNS amplification abuse.

1.7 Scan Type Comparison — -sT, -sW, -sA, -sM

Nmap supports multiple TCP scan methods. Each has different characteristics for stealth, reliability, and the information returned.

Commands

```
nmap -sT 127.0.0.1
nmap -sW 127.0.0.1
nmap -sA 127.0.0.1
nmap -sM 127.0.0.1
```



```
root@kali: ~
Session Actions Edit View Help
root@kali)~# nmap -sT 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-07 22:08 SAST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000070s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (conn-refused)

Nmap done: 1 IP address (1 host up) scanned in 0.05 seconds

root@kali)~# nmap -sW 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-07 22:08 SAST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000070s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.12 seconds

root@kali)~# nmap -sA 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-07 22:09 SAST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000070s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.10 seconds

root@kali)~# nmap -sM 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-07 22:09 SAST
```

Figure 9: TCP Connect, Window, ACK, and Maimon scan types demonstrated

- -sT (TCP Connect): Completes the full handshake. More detectable, does not need raw socket privileges. Result: 1,000 closed (conn-refused).
- -sW (Window scan): Analyses the TCP window size in RST responses to infer port state. Result: 1,000 closed (reset).
- -sA (ACK scan): Sends ACK packets to map firewall rules rather than detect open/closed state. Result: 1,000 unfiltered and no firewall in the path.

- -sM (Maimon scan): Sends FIN/ACK probes to bypass some packet filters. Behaviour is OS-dependent.

Security implication: Defensive monitoring that only watches for completed TCP connections will miss SYN, ACK, and FIN-based scans entirely. IDS rules must cover all scan types.

1.8 External Scan — scanme.nmap.org

Nmap provides scanme.nmap.org (45.33.32.156) as a publicly authorised test target.

Command

```
nmap -A -T4 scanme.nmap.org
```

```

root@kali: ~
Session Actions Edit View Help

root@kali:~# nmap -A -T4 scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-07 22:05 SAST
Warning: 45.33.32.156 giving up on port because retransmission cap hit (6).
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.077s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
22/tcp    open  ssh            OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|_ 1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|_ 2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|_ 256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_ 256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
80/tcp    open  http          Apache httpd 2.4.7 ((Ubuntu))
|_ http-server-header: Apache/2.4.7 (Ubuntu)
9929/tcp  open  nping-echo    Nping echo
31337/tcp open  tcpwrapped
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (91%), Slirp (91%), QEMU (91%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (91%), QEMU user mode network gateway (91%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 0.70 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .

```

Figure 10: Aggressive scan of scanme.nmap.org — 4 open ports found

- 22/tcp — OpenSSH 6.6.1p1 (Ubuntu). SSH host keys enumerated: DSA, RSA, ECDSA, ED25519.
- 80/tcp — Apache httpd 2.4.7 (Ubuntu).
- 9929/tcp — Nping Echo service (Nmap test service).
- 31337/tcp — tcpwrapped (service identity hidden).
- OS detected as Linux. Traceroute: 1 hop at 0.70 ms.

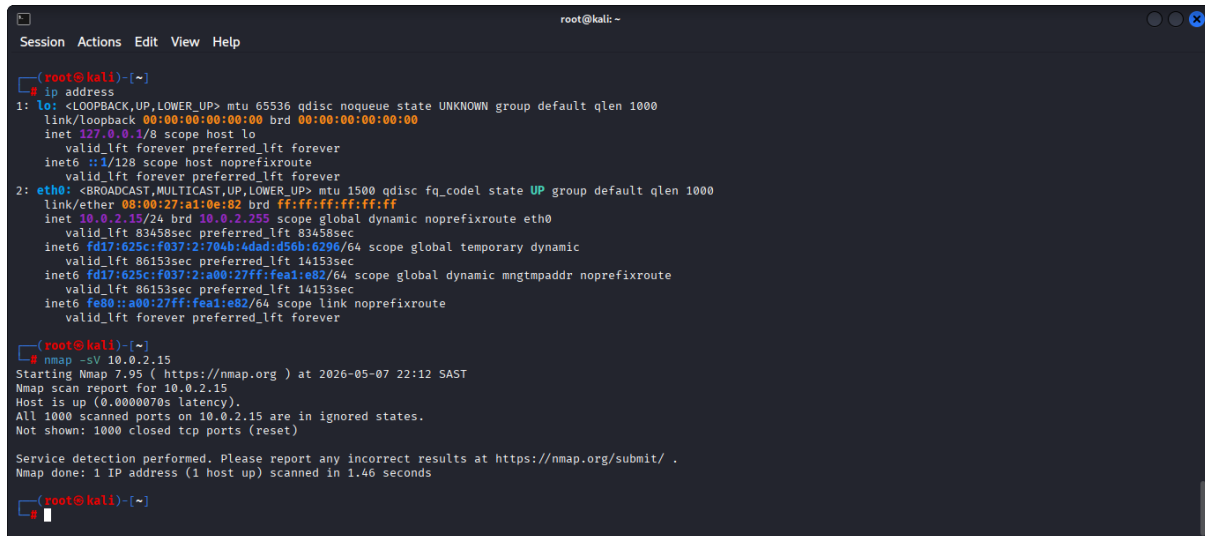
Security implication: OpenSSH 6.6.1p1 was released in 2014 and carries several known CVEs. The server should be updated. Port 31337 (historically called "elite" in hacker culture) presenting as tcpwrapped warrants further investigation.

1.9 Version Detection — nmap -sV

The -sV flag probes open ports to determine the exact software name and version running behind each service.

Command

```
ip address
nmap -sV 10.0.2.15
```



```
root@kali: ~
Session Actions Edit View Help

root@kali)~# ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a1:0e:82 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 83458sec preferred_lft 83458sec
    inet6 fd17:625c:f037:2:704b:4dad:d56b:6296/64 scope global temporary dynamic
        valid_lft 86153sec preferred_lft 14153sec
    inet6 fd17:625c:f037:2:a00:27ff:fea1:e82/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86153sec preferred_lft 14153sec
    inet6 fe80::a00:27ff:fea1:e82/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

root@kali)~# nmap -sV 10.0.2.15
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-07 22:12 SAST
Nmap scan report for 10.0.2.15
Host is up (0.0000070s latency).
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.46 seconds

root@kali)~#
```

Figure 11: ip address confirmation and -sV scan against 10.0.2.15

Result: No services detected on 10.0.2.15 (our own machine). The ip address output confirms the eth0 address as 10.0.2.15. Service detection was performed and completed in 1.46 seconds.

Security implication: Version detection is critical in real audits. Exact version numbers let testers cross-reference CVE databases instantly. Services should be configured to suppress or falsify banner information where possible.